

Bias and Fairness Analysis

Overall Survival in Acute Myeloid Leukemia

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Abstract

A global C-index of 0.752 describes a model's average behaviour but says nothing about *who* it serves well. This report audits a Random Survival Forest trained on 3,323 AML (Acute Myeloid Leukemia) patients across 10 European centres, examining whether predictive quality is distributed equitably across patient subgroups. Two independent methods establish that the model does not rely on the treating hospital. However, calibration analysis reveals a systematic and reproducible optimism bias for adverse-risk cytogenetic groups, invisible to discrimination metrics.

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1 Introduction

A global performance figure describes a model’s *average* behaviour. It says nothing about **who** the model serves well and who it serves poorly. A model reaching a C-index of 0.752 overall may be excellent for the majority of patients and unreliable for a specific and potentially vulnerable subgroup.

Bias analysis addresses a question that aggregate metrics cannot: **is predictive quality distributed equitably across patient subgroups, or does the model systematically disadvantage an identifiable population?**

In a clinical context, this question is not merely statistical. A prognostic tool that overestimates survival for high-risk patients could contribute to under-treatment, delayed referral to intensive care pathways, or inappropriate reassurance. *The direction of an error matters as much as its magnitude.*

2 A taxonomy of bias

Bias in a machine learning pipeline can originate at four distinct stages. Distinguishing them matters because each requires a different remedy.

2.1 Data-level bias

Introduced before any model is trained.

Type	Definition	Typical consequence
Sampling / representation	Some subgroups are under-represented relative to the target population	The model learns those groups poorly
Selection	Inclusion criteria systematically exclude a category of patients	Predictions do not generalise beyond the recruited population
Survivorship	Individuals who experienced the outcome early are absent by construction	Survival is structurally overestimated
Measurement	A variable is recorded differently across sources or periods	The model learns measurement artefacts rather than biology
Annotation / documentation	Missing data is concentrated in specific groups rather than random	Missingness itself becomes an informative and unfair signal

2.2 Model-level bias

Arises from the learning algorithm itself, even on well-balanced data.

Type	Definition	Typical consequence
Calibration	Predicted probabilities deviate systematically from observed frequencies	Correct ranking, but wrong absolute risk
Regression to the mean	Ensemble averaging pulls extreme predictions toward the distribution centre	Severe and mild cases are both under-differentiated

Type	Definition	Typical consequence
Discrimination	Ranking quality differs across subgroups	The model is less useful for some patients
Aggregation	A single model is fitted where subgroups follow different underlying relationships	Neither subgroup is optimally served

2.3 Leakage and confounding

Type	Definition	Typical consequence
Contextual leakage	A technical or administrative variable carries outcome signal	Performance does not transfer to a new setting
Institutional	The model relies on the treating centre rather than patient characteristics	Prognosis depends on place of care, not biology
Proxy	A neutral variable encodes a protected or structural attribute	Indirect discrimination

2.4 Deployment bias

Type	Definition	Typical consequence
Distribution shift	The deployment population differs from the training population	Silent degradation of performance
Out-of-distribution input	The interface allows inputs never observed in training	Predictions produced with no empirical basis
Automation	Users over-trust model output	Amplification of any residual bias

3 Why survival models require a specific fairness framework

Standard fairness metrics demographic parity, equalised odds, equal opportunity assume a **classification** setting with fully observed binary outcomes. Neither assumption holds here.

The outcome is a duration, not a label. There is no confusion matrix, therefore no false positive or false negative rate to equalise across groups.

The outcome is right-censored for approximately 48% of patients. For nearly half the cohort we know only that the patient was alive at last follow-up. This has two methodological consequences:

- **The C-index depends on the censoring distribution.** A subgroup with short follow-up produces fewer comparable pairs, and therefore an unstable estimate. Raw differences between groups cannot be interpreted as bias without controlling for this.
- **Kaplan–Meier is the only unbiased reference** for observed survival, and its reliability degrades as censoring increases.

Discrimination and calibration are independent failure modes. A model may rank patients correctly within a subgroup while being systematically wrong about their absolute survival. Clinically, the second error is the more dangerous, and it is entirely invisible to the C-index. Any credible fairness audit of a survival model must therefore evaluate **both**.

4 Application to our study

4.1 Available axes

The dataset contains: ID, CENTER, BM_BLAST, WBC, ANC, MONOCYTES, HB, PLT, CYTOGENETICS, CYTO_CLASS, OS_YEARS, OS_STATUS, plus patient-level molecular aggregates.

No age and no sex variable is available. Demographic fairness — the axis regulators and reviewers examine first — therefore **cannot be assessed**. This is stated as a limitation rather than worked around. It is also a modelling weakness in its own right: age is the second strongest prognostic factor in AML after cytogenetics.

Three axes remain analysable:

Axis	Levels	Fairness question
CENTER	23 hospital centres	Does prognosis depend on where the patient was treated?
CYTO_CLASS	8 ELN 2022 risk groups	Is the model as reliable for adverse-risk as for favourable-risk patients?
HAS_MOL_DATA	tested / not tested	Does the model underserve patients without molecular profiling?

CENTER is the most sensitive axis, because it is **a feature of the model itself**. The RSF could have learned institutional differences in care, recruitment or documentation rather than patient biology, a textbook case of institutional bias.

4.2 Confounders identified before analysis

Censoring is highly heterogeneous across centres , ranging from 8.4% to 90.2%, and correlates with apparent survival:

Centre	n	Censored	KM $S(2y)$	Max follow-up
MUV	83	8.4%	0.446	12.79 y
UOB	86	18.6%	0.581	14.70 y
CCH	140	76.4%	0.834	9.32 y
ROM	102	90.2%	0.809	2.84 y

Centres with short follow-up appear to have better outcomes simply because fewer deaths were observed. This is a censoring artefact, not a clinical difference.

Subgroup sizes are severely imbalanced , from 786 patients (KI) to 6 (REL), and from 1,780 (Normal) to 79 (Complex). Estimates below roughly 50 patients are statistically uninterpretable.

5 Methodology

5.1 Discrimination per subgroup

C-index computed **within** each subgroup, with **bootstrap 95% confidence intervals** (200 resamples). Subgroups below 50 patients or 15 events are reported but not interpreted.

Decision rule: bias is concluded only where confidence intervals are **disjoint** never from raw gaps. With subgroup sizes ranging over two orders of magnitude, overlapping intervals are the expected outcome even under perfect fairness.

5.2 Calibration per subgroup

Mean predicted survival at 2 years compared against the **Kaplan–Meier** estimate observed in the same subgroup:

$$\text{gap} = \overline{\hat{S}(2y)} - S_{\text{KM}}(2y) \quad (1)$$

A **positive gap means the model is too optimistic** for that group. Censoring rate is reported alongside each gap, since a heavily censored subgroup yields an unreliable KM reference.

5.3 Dependence on institutional variables

Permutation test: the **CENTER** column is randomly shuffled and the loss in C-index measured (5 repetitions). If the model genuinely relies on the treating hospital, destroying that information should degrade performance. A drop below ~ 0.005 indicates negligible dependence. This is cross-checked against the SHAP analysis conducted independently at the individual patient level.

5.4 Overfitting control

Because subgroup metrics are computed on training data, the calibration analysis is repeated on a held-out split (25%, stratified on event status) to verify that any observed miscalibration is not an artefact of memorisation.

6 Results

6.1 Discrimination — no detectable bias

Axis	Observed range	Evaluable	Conclusion
CENTER	0.752 – 0.896	11 / 23	all CIs overlap → no bias
CYTO_CLASS	0.743 – 0.817	7 / 8	weak, non-significant trend
HAS_MOL_DATA	0.798 vs 0.810	2 / 2	no access-related inequity

Selected centre results:

Even the extremes (CGM at 0.896, RMCN at 0.752) are not cleanly separated once uncertainty is accounted for. **The model ranks patients with comparable effectiveness across all evaluable subgroups.**

By cytogenetic class, a mild gradient appears, **Normal / 5q deletion** around 0.80–0.82 against **Monosomy 7** (0.743) and **Complex** (0.747) but intervals overlap. The most plausible interpretation is structural rather than a model defect: in very-poor-prognosis groups most patients die quickly, leaving less to discriminate.

Centre	n	events	C-index	95% CI
CGM	107	37	0.896	[0.848 – 0.932]
PV	316	81	0.861	[0.828 – 0.898]
DUS	445	288	0.807	[0.784 – 0.832]
KI	786	483	0.800	[0.781 – 0.820]
RMCN	182	105	0.752	[0.707 – 0.802]

6.2 Institutional dependence — ruled out

Test	Result
C-index (reference)	0.8105
Drop when CENTER is shuffled	0.0051 ± 0.0002
Relative impact	0.6% of performance

Destroying all information about the treating hospital costs the model 0.5% of a C-index point. This is corroborated independently by the SHAP analysis, where *Treatment center* contributes ≈ 0 to individual predictions.

Prognosis does not depend on where the patient was treated. This is a negative result, and a strong one: the variable is present in the model but effectively inert.

6.3 Calibration — systematic bias on adverse-risk groups

This is the principal finding.

Group	n	Predicted	Observed	Gap	Held-out	Censored
Complex	79	0.425	0.269	+0.156	+0.148	27.8%
Monosomy 7	164	0.345	0.246	+0.099	+0.102	25.0%
Trisomy 8	181	0.479	0.428	+0.052	+0.021	35.9%
Missing	285	0.600	0.591	+0.009	+0.004	40.4%
5q deletion	296	0.567	0.560	+0.007	−0.051	41.2%
Other	304	0.599	0.608	−0.010	+0.016	48.7%
Normal	1780	0.679	0.699	−0.020	−0.024	57.2%

The model is systematically too optimistic for patients with adverse cytogenetics. For complex karyotypes it predicts 42.5% survival at two years where 26.9% is observed a 15.6-point overestimation.

Three elements establish this as a genuine model bias rather than noise:

1. **The gradient is monotonic.** The worse the prognosis, the larger the overestimation. Favourable and majority groups are well calibrated ($|\text{gap}| < 0.02$).
2. **It reproduces out of sample.** Held-out gaps (+0.148, +0.102) match training gaps (+0.156, +0.099) almost exactly, excluding overfitting.
3. **It is not a censoring artefact.** Monosomy 7 has only 14.7% censoring in the held-out split, so its Kaplan–Meier reference is reliable.

Mechanism. This is **regression to the mean**, an expected property of random forests: each prediction aggregates leaf averages, pulling extreme cases toward the centre of the outcome

distribution. The affected groups are simultaneously the most severe and the least represented (79 and 164 patients), which amplifies the effect. The bias therefore *originates in the data* (under-representation of severe cases) and *manifests in the model* (shrinkage of their predicted risk).

Clinical implication. The error runs in the most harmful direction: announcing a more favourable prognosis than reality to the highest-risk patients. It is entirely invisible to the C-index. `Complex` retains a respectable discrimination of 0.747 and surfaces only through calibration analysis. This validates the two-pronged methodology.

7 Bias inventory

7.1 Confirmed

Bias	Evidence	Origin
Calibration / regression to the mean	+0.156 (<code>Complex</code>), +0.099 (<code>Monosomy 7</code>), reproduced out-of-sample	Model, driven by data imbalance
Representation	786 $\rightarrow$ 6 patients per centre; 1,780 $\rightarrow$ 79 per cytogenetic class	Data
Censoring	8.4% $\rightarrow$ 90.2% across centres, correlated with apparent survival	Data / study design
Documentation	285 patients with <code>CYTO_CLASS = Missing</code> , 256 without molecular testing	Data collection
Contextual leakage (suspected)	<code>DEPTH_MEAN</code> / <code>DEPTH_MAX</code> carry prognostic signal despite being purely technical	Feature engineering

7.2 Investigated and ruled out

Bias	Evidence
Institutional	<code>CENTER</code> permutation costs 0.0051 C-index; SHAP contribution ≈ 0
Discrimination	All subgroup confidence intervals overlap
Access-to-care inequity	<code>HAS_MOL_DATA</code> : 0.798 vs 0.810, overlapping intervals

7.3 Not assessable

Bias	Reason
Demographic	Neither age nor sex present in the dataset
Selection	No information on excluded patients or referral patterns
Survivorship	Patients dying before karyotyping are absent by construction

8 Discussion

The analysis produces one strong negative result and one strong positive result, and the contrast between them is instructive.

The negative result is reassuring. Despite `CENTER` being an explicit model feature (a configuration that would normally raise concern) the model does not exploit it. Two independent methods (permutation testing at the population level, SHAP at the individual level) converge on the same conclusion. The model bases its prognosis on biology, not on institutional context.

The positive result is actionable. The calibration bias is systematic, reproducible, and runs in the clinically harmful direction. Critically, it would have gone entirely unnoticed had the audit been limited to discrimination metrics: `Complex` shows a C-index of 0.747, which any standard evaluation would deem acceptable.

This illustrates the central methodological point of the report: **for survival models, ranking quality and absolute risk accuracy are separate properties, and fairness must be assessed on both.** A model can rank a subgroup perfectly while being systematically wrong about how long its patients will actually live.

The mechanism also clarifies where remediation should occur. Since the bias stems from under-representation of severe cases combined with ensemble averaging, changing algorithm is unlikely to resolve it. Post-hoc recalibration or reweighting of severe cases are the appropriate directions.

9 Limitations

- **Metrics are computed on training data.** Subgroup C-indices around 0.80 correspond to the training C-index (0.8105), not to the held-out platform score (0.752). Absolute values are optimistic; only *relative comparisons between subgroups* are interpretable.
- **The held-out split is imperfect.** It is drawn from patients the model saw during training. A cross-validated calibration, measured on each validation fold, would be required for publication-grade evidence.
- **Small subgroups.** 12 of 23 centres fall below the interpretability threshold. Held-out `Complex` and `Monosomy 7` rest on 22 and 34 patients respectively (indicative results requiring confirmation on an external cohort).
- **No demographic variables,** so the most commonly examined fairness axis is inaccessible.
- **A single horizon** (2 years) was used for calibration. Bias may behave differently at 1 or 5 years.
- **Out-of-distribution inputs in the application.** The Streamlit interface offers cytogenetic classes absent from training (`inv(16)`, `t(8;21)`) or represented by a single patient (`APL t(15;17)`), which would produce predictions with no empirical basis. This is a deployment bias identified during the audit and requiring correction before any use.